

AI Investment Simulator

36-Portal Learning Syllabus

PURPLE -> BLUE -> GREEN -> YELLOW

Learn investing by actually investing.

Course Concept

The learner begins with \$14,800 in virtual cash and develops a portfolio while progressing through four investment cities. Each city contains nine syllabus contact portals. New knowledge is applied directly to the persistent Portfolio Lab, allowing the learner to research opportunities, buy assets, add to positions, hold, trim, exit, hedge and rebalance as markets change.

- 36 syllabus contact portals: 9 per city.
- Four progression stages: Purple, Blue, Green and Yellow.
- Purple builds foundations; Blue develops research and asset selection; Green develops portfolio management; Yellow develops professional mastery.
- Investment universe includes stocks, whole-company acquisitions, real estate, forex, bonds, ETFs, commodities, alternatives and cash.
- Portfolio performance is judged alongside research quality, risk management and decision process - not returns alone.

PURPLE CITY - FOUNDATION

Portals 1-9 | Learn how markets and investing work

Portal	Topic	Core Development	Portfolio Application
1	Enter the Market	Investing vs saving vs trading; risk and return; compound growth; inflation and real returns.	Receive \$14,800 in virtual cash and establish the first investment objective.
2	Know Your Investor	Risk tolerance, risk capacity, investment horizon, liquidity needs and return objectives.	Create a personal investment mandate.
3	Understanding Asset Classes	Stocks, bonds, ETFs, real estate, businesses, forex, commodities and cash.	Build an initial asset-class watchlist.
4	How Markets Work	Exchanges, brokers, bid/ask prices, liquidity, market and limit orders, and transaction costs.	Execute the first simulated transaction.
5	Reading the Market	Price charts, market capitalization, indices, trading volume, volatility and sentiment.	Interpret basic market information before making a decision.
6	Research Fundamentals	Company reports, economic data, industry research, news, source credibility and research logs.	Research several opportunities and rank the quality of the evidence.
7	Build Your First Portfolio	Position sizing, portfolio	Deploy part or all of the

		weights, diversification and cash allocation.	\$14,800 into a defensible starting portfolio.
8	Investment Decisions	Buy, add, hold, trim and exit decisions; opportunity cost and decision rationale.	Make and document active portfolio decisions.
9	Purple City Challenge	Review of foundations, holdings, allocation, evidence and portfolio weaknesses.	Defend the initial portfolio and rebalance before entering Blue City.

BLUE CITY - ANALYSIS & ASSET SELECTION

Portals 10-18 | Learn how to find and value investments

Portal	Topic	Core Development	Portfolio Application
10	Understanding Companies	Business models, revenue drivers, customers, competitors, market share and competitive advantage.	Compare companies before selecting one for deeper analysis.
11	Financial Statements	Income statement, balance sheet, cash-flow statement, working capital, debt and free cash flow.	Assess how money moves through a company.
12	Financial Health	Margins, profitability, leverage, liquidity, ROE, ROIC and free-cash-flow quality.	Identify strengths, weaknesses and financial red flags.
13	Company Valuation	P/E, PEG, P/B, EV/EBITDA, free-cash-flow yield, DCF and margin of safety.	Estimate value and determine a defensible purchase price.
14	Stock Selection	Growth, value, quality, dividend, momentum, contrarian and GARP approaches.	Select stocks and determine appropriate position sizes.
15	Buying an Entire Company	Private-business valuation, enterprise value, normalized earnings, due diligence, acquisition financing and deal risk.	Evaluate and negotiate a simulated whole-company acquisition.
16	Real Estate Investing	Rental income, NOI, cap rates, vacancy, financing, leverage, cash-on-cash return, appreciation and REITs.	Underwrite a property and compare it with liquid investment alternatives.
17	Forex Investing	Currency pairs, exchange rates, pips, spreads, interest-rate differentials, central banks, macro drivers and hedging.	Research and size a simulated currency position.
18	Blue City Investment Committee	Cross-asset comparison of stocks, business acquisition, property and forex.	Allocate capital among competing opportunities and defend the choices.

GREEN CITY - PORTFOLIO MANAGEMENT

Portals 19-27 | Learn how to manage investments as conditions change

Portal	Topic	Core Development	Portfolio Application
19	Bonds & Fixed Income	Government and corporate bonds, yields, duration, credit risk and interest-rate sensitivity.	Decide whether fixed income improves the portfolio.

20	ETFs, Commodities & Alternatives	Index funds, sector ETFs, gold, commodities and alternative exposures.	Compare direct ownership with pooled and alternative exposures.
21	Strategic Asset Allocation	Long-term allocation across stocks, businesses, property, forex, bonds, alternatives and cash.	Create a target multi-asset portfolio.
22	Portfolio Diversification	Correlation, concentration, sector, geographic and currency exposure, and liquidity.	Identify hidden concentrations and diversification gaps.
23	Add, Trim or Exit	Rebalancing bands, valuation discipline, thesis deterioration, sell rules and capital rotation.	For every holding choose: ADD, HOLD, TRIM, EXIT or REPLACE.
24	Macroeconomics for Investors	GDP, inflation, rates, unemployment, monetary and fiscal policy, yield curves and economic cycles.	Assess implications across every major portfolio asset class.
25	Earnings, News & New Information	Earnings, guidance, analyst revisions, regulation, rate announcements and property-market information.	Distinguish thesis-changing evidence from market noise.
26	Risk Management & Stress Testing	Volatility, beta, drawdown, Sharpe ratio, correlation, VaR, leverage, liquidity and scenario analysis.	Stress-test the portfolio against recession, inflation, property, currency and company shocks.
27	Green City Crisis Challenge	Integrated market-shock simulation across multiple holdings and asset classes.	Decide what to sell, protect, add to or leave untouched under pressure.

YELLOW CITY - MASTER INVESTOR

Portals 28-36 | Professional portfolio construction and optimization

Portal	Topic	Core Development	Portfolio Application
28	Build an Investment Thesis	Expected return, catalysts, risks, variant perception, falsification and sell criteria.	Create formal theses for major holdings.
29	Behavioural Finance	Loss aversion, FOMO, anchoring, overconfidence, herding, confirmation bias and disposition effect.	Use the learner's trade history to identify behavioural weaknesses.
30	Portfolio Optimization	Expected return, risk-adjusted performance, correlations, concentration, risk budgets and capital efficiency.	Propose an improved portfolio subject to realistic constraints.
31	Capital Rotation	Opportunity cost, relative attractiveness, valuation changes and new investment opportunities.	Decide where capital should remain and where it should be redeployed.
32	Historical Market Simulator	Market environments resembling 1987, dot-com, 2008, COVID and the 2022 inflation/rate shock.	Manage without knowing the historical outcome in advance.
33	Global Investment Strategy	International markets, country risk, emerging/developed markets, currency exposure and geopolitical risk.	Develop a global allocation and manage cross-border risks.
34	The Fund Mandate	Investment Policy Statement, return target, risk limits, liquidity, benchmarks, concentration and allocation	Graduate from the \$14,800 portfolio to a simulated \$10 million professional fund.

		constraints.	
35	Multi-Year Fund Simulation	Company events, acquisitions, property changes, recessions, currencies, rates, sectors and new opportunities.	Continuously research, buy, add, hold, trim, sell, hedge and rebalance through several simulated years.
36	Final Investment Committee	Performance attribution, benchmark comparison, drawdown, asset-class decisions, optimization, mistakes, skill vs luck and investment philosophy.	Defend the final fund before an AI Investment Committee.

Persistent Portfolio Lab

The Portfolio Lab operates alongside all 36 portals. Progress through the syllabus does not reset the portfolio. Instead, every new competency expands the learner's ability to evaluate and manage the same evolving pool of capital.

- BUY - open a new investment when the expected reward justifies the risk.
- ADD - increase a position when evidence strengthens or valuation becomes more attractive.
- HOLD - retain an investment deliberately when the thesis remains intact.
- TRIM - reduce exposure because of valuation, concentration, risk or a superior opportunity.
- EXIT - remove an investment when the thesis breaks or capital has a better use.
- HEDGE - reduce a defined portfolio risk where appropriate.
- REBALANCE - restore or deliberately change target portfolio exposures.

Assessment Philosophy

The simulator rewards disciplined investing rather than lucky outcomes. Learners are evaluated on research, valuation, portfolio construction, risk management, active management, behavioural discipline, decision quality and performance.

- Research quality and source credibility
- Valuation and underwriting
- Portfolio construction and diversification
- Quality of buy/add/hold/trim/exit decisions
- Risk management and stress testing
- Behavioural discipline and willingness to revise a thesis
- Risk-adjusted performance and benchmark comparison
- Clarity of reasoning and decision journal

Four-City Progression

City	Portals	Development Stage	Learner Progression
Purple	1-9	Foundation	Learn how to invest.
Blue	10-18	Analysis & Selection	Learn what to invest in.
Green	19-27	Portfolio Management	Learn how to manage and adapt.
Yellow	28-36	Master Investor	Prove you can manage

			capital professionally.
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Core Learning Loop

LEARN -> RESEARCH -> DECIDE -> ALLOCATE -> EXPERIENCE -> MEASURE -> REFLECT ->
ADAPT